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Math 208 Major Exam II Term 252

Original exam date: 13 April 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $W(x)$ is the Wronskian of the functions

$$f_1(x) = e^x, \quad f_2(x) = \cos(2x), \quad \text{and} \quad f_3(x) = \sin(2x),$$

then $W(0) =$

(a) 0

(b) -10

(c) -8

(d) 10

(e) 8

2. The general solution of the differential equation

$$y^{(4)} - 4y''' + 3y'' + 4y' - 4y = 0$$

is

(a) $y = c_1e^x + c_2e^{-2x} + (c_3 + c_4x)e^{2x}$

(b) $y = c_1e^x + c_2e^{-x} + c_3e^{2x} + c_4e^{4x}$

(c) $y = c_1e^x + c_2e^{-x} + (c_3 + c_4x)e^{2x}$

(d) $y = c_1e^{-x} + c_2e^{2x} + (c_3 + c_4x)e^x$

(e) $y = c_1e^x + c_2e^{2x} + (c_3 + c_4x)e^{-x}$

3. The characteristic equation of the matrix

$$\begin{bmatrix} 0 & -2 & -1 \\ -3 & -1 & -3 \\ 2 & 2 & -3 \end{bmatrix}$$

is

(a) $\lambda^3 + 3\lambda^2 - \lambda - 2 = 0$

(b) $\lambda^3 - 2\lambda^2 - \lambda - 2 = 0$

(c) $\lambda^3 + 4\lambda^2 + 5\lambda - 34 = 0$

(d) $\lambda^3 - 4\lambda^2 + 5\lambda - 34 = 0$

(e) $\lambda^3 + 4\lambda^2 - 5\lambda + 34 = 0$

4. The solution of the initial value problem

$$y'' + y = 0, \qquad y(0) = 3, \quad y'(0) = 0$$

is given by

(a) $y = 3\cos x + 2\sin x$

(b) $y = 3\cos x + \sin x$

(c) $y = 3\sin x + 3$

(d) $y = 3\cos x - \sin x$

(e) $y = 3\cos x$

5. Which pair of vectors in $\mathbb{R}^3$ are linearly independent?

(a) $v_1 = (3, 5, 2)$, $v_2 = \left(\frac{3}{4}, \frac{5}{4}, \frac{1}{2}\right)$

(b) $v_1 = (2, 1, 4)$, $v_2 = (4, 2, 5)$

(c) $v_1 = (1, 0, 5)$, $v_2 = \left(\frac{1}{3}, 0, \frac{5}{3}\right)$

(d) $v_1 = (0, 0, 0)$, $v_2 = (1, 1, 1)$

(e) $v_1 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{3}\right)$, $v_2 = \left(\frac{1}{4}, \frac{1}{8}, \frac{1}{6}\right)$

6. A particular solution of the differential equation

$$y'' - 4y' + 4y = x^{-1}e^{2x}$$

is given by $y_p(x) =$

(a) $xe^{3x}\ln|x|$

(b) $x^2e^{2x}\ln|x|$

(c) $xe^{2x}\ln|x|$

(d) $xe^{-2x}\ln|x|$

(e) $x^2e^{-2x}\ln|x|$

7. A particular solution of the differential equation

$$y'' - 3y' + 2y = \cos\left(e^{-x}\right)$$

is given by $y_p(x) =$

(a) $-e^{2x}\cos\left(e^{-x}\right)$

(b) $e^{x}\cos\left(e^{-x}\right)$

(c) $e^{-2x}\cos\left(e^{-x}\right)$

(d) $e^{2x}\cos\left(e^{x}\right)$

(e) $-e^{x}\cos\left(e^{-x}\right)$

8. If the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix $D = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$ such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 2 & 1 \\ -1 & -2 \end{bmatrix}$

(e) $P = \begin{bmatrix} 2 & -1 \\ 1 & -2 \end{bmatrix}$

9. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$(D+2)^3(D^2+4)y = e^{-2x} - \sin(2x)$$

is given by $y_p(x) =$

(a) $Ax^3e^{-2x} + Bx\sin(4x) + Cx\sin(4x)$

(b) $Axe^{-2x} + Bx\sin(2x) + Cx\cos(2x)$

(c) $Ax^2e^{-2x} + Bx\sin(2x) + Cx\cos(2x)$

(d) $Ax^3e^{-2x} + Bx\sin(2x) + Cx\cos(2x)$

(e) $Ax^3e^{-2x} + B\sin(2x) + Cx\cos(2x)$

10. The rank of the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ -1 & 1 & 2 & 0 \\ 2 & 1 & -3 & 1 \\ 1 & 5 & 0 & 6 \end{bmatrix}$$

is

(a) 1

(b) 2

(c) 0

(d) 4

(e) 3

11. Given that $y = \cos(2x)$ is a solution of the differential equation

$$y^{(4)} + 4y''' + 8y'' + 16y' + 16y = 0.$$

The general solution of the differential equation is

(a) $y = c_1 e^{-2x} + c_2 e^{-x} + c_3 \cos(2x) + c_4 \sin(2x)$

(b) $y = c_1 e^{-2x} + c_2 e^{2x} + c_3 \cos(2x) + c_4 \sin(2x)$

(c) $y = (c_1 + c_2 x)e^{-2x} + c_3 x \cos(2x) + c_4 \sin(2x)$

(d) $y = (c_1 + c_2 x)e^{-2x} + c_3 \cos(2x) + c_4 \sin(2x)$

(e) $y = (c_1 + c_2 x)\cos(2x) + (c_3 + c_4 x)\sin(2x)$

12. Given the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$. A basis for the eigenspace that corresponds to the eigenvalue $\lambda = 2$ is $v_1 = \begin{bmatrix} 0 \\ \alpha \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ \beta \\ 0 \end{bmatrix}$, where $2\alpha + \beta =$

(a) 2

(b) -2

(c) 1

(d) 4

(e) 3

13. If $y_p = A + Bx + Cx^2 + De^x$ is a particular solution of the differential equation

$$y'' - 6y' + 9y = 27x^2 + 4e^x,$$

then $A + B + C + D =$

(a) 0

(b) 10

(c) 11

(d) 8

(e) 12

14. A linear homogeneous differential equation with real coefficients, having the solutions:

$$2, \quad 3x\, e^x \cos(3x)$$

is

(a) $D(D^2 + 2D + 10)^2\, y = 0$

(b) $D(D^2 - 2D + 10)^2\, y = 0$

(c) $D(D^2 - 2D + 10)\, y = 0$

(d) $D(D^2 - D + 8)^2\, y = 0$

(e) $D(D^2 + 2D + 10)\, y = 0$

15. If the solution

$$\begin{cases} x_1 + 3x_2 - 2x_3 + x_4 = 0 \\ 2x_1 + 6x_2 - x_3 - 7x_4 = 0 \\ x_1 + 3x_2 + x_3 - 8x_4 = 0 \end{cases}$$

consists of all linear combinations of the vectors: $v_1 = (a, b, 0, 0)$ and $v_2 = (c, 0, 3, d)$, then $\dfrac{a}{b} + \dfrac{c}{d} =$

(a) 0

(b) -3

(c) 3

(d) -2

(e) 2

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$W(0) = \det \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 1 & -4 & 0 \end{bmatrix} = 8 + 2 = 10.$
2	**(c)**	$(r-1)(r+1)(r-2)^2 = 0$, so $y = c_1 e^x + c_2 e^{-x} + (c_3 + c_4 x)e^{2x}$.
3	**(c)**	$\det(\lambda I - A) = \lambda(\lambda^2 + 4\lambda + 9) - (6\lambda + 30) + (2\lambda - 4)$.
4	**(e)**	$r = \pm i$: $y = c_1 \cos x + c_2 \sin x$, $c_1 = 3$, $c_2 = 0$, so $y = 3\cos x$.
5	**(b)**	$(4, 2, 5) \neq k(2, 1, 4)$ since $\frac{4}{2} = \frac{2}{1} \neq \frac{5}{4}$; every other pair is a multiple or has $\mathbf{0}$.
6	**(c)**	Var. of parameters, $W = e^{4x}$: $u_1 = -x$, $u_2 = \ln\lvert x\rvert$, so $y_p = xe^{2x}\ln\lvert x\rvert$.
7	**(a)**	Var. of parameters, $W = e^{3x}$, $u = e^{-x}$: $y_p = -e^{2x}\cos(e^{-x})$.
8	**(c)**	$\lambda = 2$: $(2, 1)$; $\lambda = -3$: $(1, -2)$, so $P = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$.
9	**(d)**	$r = -2$ (triple), $\pm 2i$ (simple): $y_p = Ax^3e^{-2x} + x(B\sin 2x + C\cos 2x)$.
10	**(e)**	Echelon form has rows $(1, 2, -1, 3), (0, 3, 1, 3), (0, 0, 0, -2), \mathbf{0}$: rank 3.
11	**(d)**	$(r^2+4)(r+2)^2 = 0$: $r = -2, -2, \pm 2i$, so $y = (c_1 + c_2 x)e^{-2x} + c_3\cos 2x + c_4 \sin 2x$.
12	**(a)**	$A - 2I$: $x_2 = x_3$, basis $(0, 1, 1), (1, 0, 0)$, so $\alpha = 1$, $\beta = 0$, $2\alpha + \beta = 2$.
13	**(b)**	$C = 3$, $B = 4$, $A = 2$, $D = 1$ (from $4De^x = 4e^x$): sum $= 10$.
14	**(b)**	$r = 0$ and $r = 1 \pm 3i$ twice: $r(r^2 - 2r + 10)^2$, i.e. $D(D^2 - 2D + 10)^2 y = 0$.
15	**(e)**	$x_3 = 3x_4$, $x_1 = -3x_2 + 5x_4$: $(-3, 1, 0, 0)$, $(5, 0, 3, 1)$, so $-3 + 5 = 2$.